

HIGHFIELD HIGH SCHOOL

Advanced level Mathematics

PURE MATHEMATICS

6042/2

3 hours

Midyear 2024 Exam Session/GM

PAPER 2

Additional Materials:

Answer Paper

List of formulae MF7

Graph paper

Scientific calculator [Non-programmable]

INSTRUCTIONS TO CANDIDATES

Write your name centre number and candidate number on the first page of your answer papers.

Answer **all** questions in Section A and any **five** questions from section B.

If a numerical answer cannot be given exactly and the accuracy required is not specified in the question, then in the case of an angle it should be given to the nearest degree, and in other cases it should be given to correct to 2 significant figure,s.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [] at the end of each question or part question.

This question paper consists of 4 printed pages

Section A (40 Marks).

Answer *all* questions in this section.

- 1 If x is small and measured in radians show that
- $$4\sin\left(x + \frac{\pi}{6}\right) - 8\cos\left(x - \frac{\pi}{6}\right) = (2\sqrt{3} - 1)x^2 + (2\sqrt{3} - 4)x + 2 - 4\sqrt{3}. \quad [6]$$
- 2 Find the constant p such that $x^2 + 2$ is a factor of $x^4 - 6x^2 + p$. [6]
- 3 (a) Solve the equation $|3^x - 5| = 1$ [4]
 (b) Solve the simultaneous equations

$$e^{2x+4} = ey.$$

$$\ln y = 4x + 6$$
 [4]
- 4 (a) A sequence is defined by $U_n = \sin\left(\frac{n\pi}{2}\right)$, for $n = 1, 2, 3, \dots$.
 Write down the first five terms of the sequence and hence describe the behaviour of the sequence as $n \rightarrow \infty$. [3]
- (b) The first term of a geometric series is 120. The sum to infinity is 480.
 Calculate the smallest value of the number of terms n such that the sum of the first n terms is greater than 300. [6]
- 5 (a) Show that $\int_0^{\ln 2} (e^x + 2e^{-x})^2 dx = 3 + 2\ln 4$. [5]
- (b) The parametric equations of a curve are $x = e^{-t} \cos t$ and $y = e^{-t} \sin t$.
 Show that $\frac{dy}{dx} = \tan\left(t - \frac{\pi}{4}\right)$. [6]

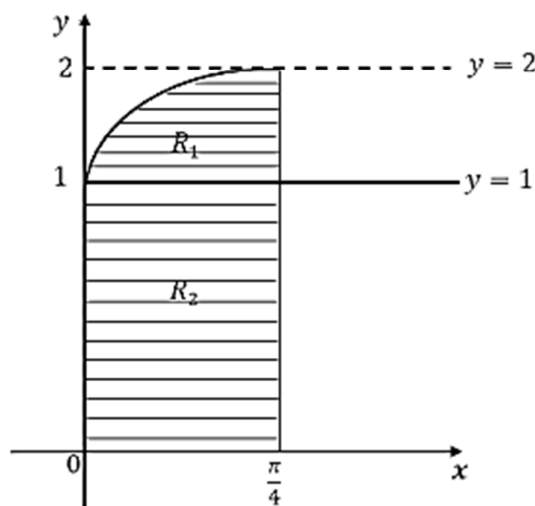
Section B (80 Marks)

Answer any *five* questions from this section. Each question carries **16** marks

- 6 (a) Solve the differential equation $\cos x \frac{dy}{dx} + 4y^2 \sin x = \sin x$, expressing y in terms of x given that $y = \frac{15}{34}$ when $x = \frac{\pi}{3}$. [7]
- (b) Find the exact coordinates of the turning point(s) on the curve
 $y = (x - 1)(x - 2) + \ln x$ and determine their nature. [6]
- (c) Show that $\frac{a^{-1}}{a + \sqrt{a}}$ can be written in the form $1 - a^y$, stating the value of y . [3]

- 7 (a) The equation of the plane π is $3x + 4y - z = 43$.
Find
- (i) the vector equation of the line through the point $A(2, 3, 1)$ and at right angles to the plane π [2]
 - (ii) the coordinates of B the point of intersection of the line in (i) and the plane π . [4]
 - (iii) the coordinates of D the reflection of A in the plane π . [4]
 - (iv) Find the coordinates of the point where the plane π intersects the z axis. [2]
- (b) Given the point C with co-ordinates $(6, -3, -3)$. Find the Cartesian equation of the plane ABC . [4]

- 8 (a) The diagram shows the regions R_1 and R_2 enclosed by the curve $y = 1 + \sin 2x$, the line $x = \frac{1}{4}\pi$ and the axes. The line $y = 1$ divides the two regions.



- Use the trapezium rule with 4 equal intervals, to find the approximate total area of $R_1 + R_2$ correct to 3 decimal places. [5]
- (b) (i) Use the graphical argument to show that the equation $e^{2x} + 4x - 5 = 0$ has only one real root. [3]
- (ii) Taking 0.5 to be the first approximation to the root, use the Newton-Raphson method 2 times to obtain a better approximation correct to 3 decimal places. [4]
- (c) A 2% error is made in measuring the radius of a sphere.
Find the percentage error in the surface area of the sphere. [4]
[Surface area of a sphere = $4\pi r^2$]

- 9 (a) Given that $3 + i$ is a root of $2z^4 - 14z^3 + 33z^2 - 26z + 10 = 0$, find the other three roots. [6]
- (b) Shade on an Argand diagram the region satisfying both the loci $|z - i| < |z - 2|$ and $\frac{\pi}{2} \leq \arg(z - 2) < \frac{2\pi}{3}$. [4]
- (c) Given that $z = \cos \theta + i \sin \theta$. Show that $z + \frac{1}{z} = 2 \cos \theta$. [2]
Hence express $4 \cos^3 \theta$ in terms of $\cos 3\theta$ and $\cos \theta$. [4]
- 10 (a) Show that the set $(\mathbb{Z}_4, +)$, under addition modulo 4 forms a group. [7]
- (b) The binary operation $*$ on the set $\mathbb{R}$ of real numbers is defined by $a * b = \frac{a+b}{5}$, where $a, b \in \mathbb{R}$.
- (i) Find $(2 * 3)$. [1]
- (ii) Solve the equation $(2 * x) * 1 = 3$ [3]
- (ii) Determine whether the binary operation has an identity element justifying your answer. [3]
- (c) Define the term **closure** with reference to a set S under a binary operation $*$. [2]
- 11 (a) Given the matrices $P = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 2 & 1 \\ 4 & 1 & 1 \end{pmatrix}$ and $PQ = \begin{pmatrix} 1 & 1 & 1 \\ 3 & 3 & 1 \\ 1 & -1 & 1 \end{pmatrix}$.
Find
- (i) the inverse of P . [6]
- (ii) the matrix Q . [3]
- (iii) the point whose image is $(1, -1, 2)$ under mapping by P . [3]
- (b) Show that the line $y = 2x - 1$ is mapped onto itself under the transformation represented by the matrix $\begin{pmatrix} 3 & -1 \\ 4 & -1 \end{pmatrix}$. [4]
- 12 (a) (i) Express in partial fractions $f(x) = \frac{x^2+8x+13}{(x+2)(x+3)}$. [4]
- (ii) Hence find the values of a , b and c if $f(x) \equiv a + bx + cx^2$. [4]
- (b) Given the matrix $A = \begin{pmatrix} 2 & 0 \\ 0 & 1 \end{pmatrix}$.
- (i) Find A^2 and A^3 . [2]
- (ii) Hence conjecture a formula for A^n . [1]
- (iii) Use mathematical induction to prove your conjecture in (b)(ii). [5]